Integrated Facility Layout And Material Handling System Design In Semiconductor Fabrication Facilities

Brett A. Peters
Taho Yang

Department of Industrial Engineering
Texas A&M University
College Station, TX 77843-3131

(409) 845-3574
(409) 847-9005
bpeters@tamu.edu

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Address correspondence to Brett A. Peters, Dept. of Industrial Engineering, Texas A&M University, College Station, TX 77843–3131 or email to bpeters@tamu.edu

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ABSTRACT

Semiconductor manufacturing is an important component of the US manufacturing industry. Most of today's fabrication facilities and those being designed for the near future use a bay layout configuration and an overhead monorail system for moving material between bays. These material handling systems are usually designed with a spine or perimeter type of configuration. This paper investigates the layout and material handling system design integration problem in semiconductor fabrication facilities and proposes a methodology for solving this integrated design problem. A spacefilling curve approach is used to address the facility layout, while the structure of the spine and perimeter configurations are exploited to create a network flow problem to determine the material handling system design. Computational results are presented and show exceptional promise for this procedure in solving the integrated design problem in a semiconductor manufacturing environment.

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I. INTRODUCTION

Semiconductor manufacturing is an important component of the US manufacturing industry. Most of today's fabrication facilities (fabs) and those being designed for the near future use a bay layout configuration [1]. In this approach, the facility is divided into a number of bays that contain processing equipment. Typically, the bays contain similar pieces of equipment, e.g., ion implanters. This situation creates a large amount of material flow between bays (interbay movement), especially since the semiconductor manufacturing process is highly reentrant.

While there may be other, perhaps more efficient layout arrangement strategies, the semiconductor industry is reluctant to adopt these strategies since the bay configuration offers many advantages in terms of maintenance and operation of the physical equipmen[t 1]. That is, since all of the equipment in a bay is of the same type, the utility distribution system can be simplified. Furthermore, the equipment used in semiconductor processing is very sensitive, due to the tight tolerances required, and hence is down for preventive maintenance and repair a large portion of the time. Having the processing equipment of a particular type collocated ensures that production can continue during these maintenance periods. Finally, having similar processing equipment collocated allows maintenance passages to be constructed between the bays for monitoring and repairing the equipment. These maintenance areas are separate from the cleanroom space and allow access to some parts of the equipment without disrupting production.

Most new fabs use an overhead monorail system for moving material between bays. This type of automated material handling system (AMHS) is usually designed with a spine or perimeter type of configuration, which forms a material flow loop within the facility. An AMHS with a spine configuration typically has one directed flow loop and crossover turntables for changing travel directions1 [,2] as shown in Fig. 1.

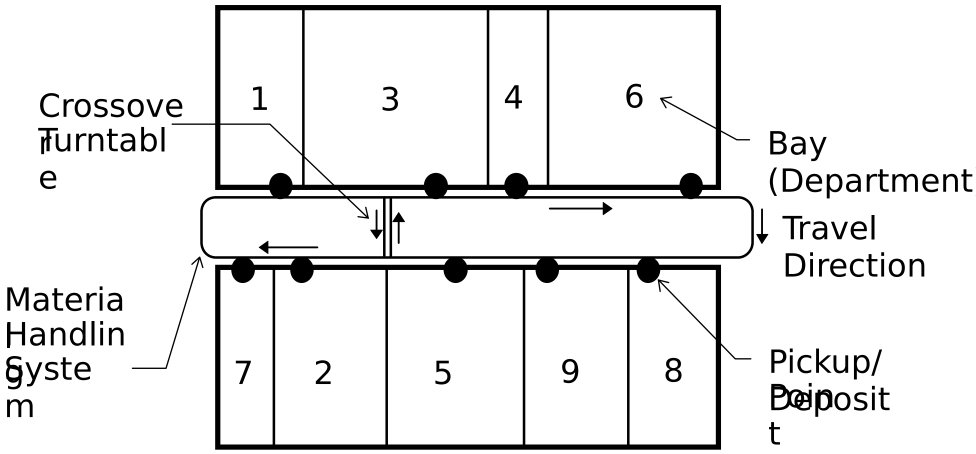


Fig. 1. Spine configuration.

An AMHS with a perimeter configuration is typically designed with two physically separate loops to form unimpeded two-way travel around the perimeter of the facility and with crossover turntables for switching travel directions [2] as shown in Fig. 2. In this configuration, each bay can contain at most two departments to ensure that each department will have at least one boundary on the perimeter to interface with the AMHS.

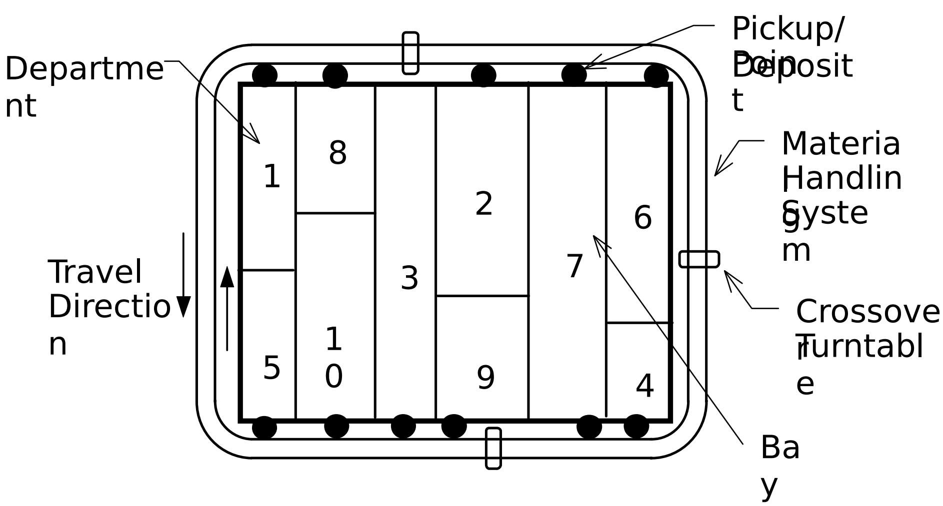


Fig. 2. Perimeter configuration.

Because of the process layout of the semiconductor fab, the interbay material handling function becomes extremely important. This material handling function provides the critical exchange between sequential fabrication process steps [3]. Furthermore, experimental data suggests that installation of an

AMHS in a semiconductor fab can significantly reduce shocks and vibrations from material handling and potentially increase yield [4].

The interactions between the facility layout and material handling system are widely recognized. The impact of these factors on the performance of the manufacturing system is also widely acknowledged5][. Hence, a good layout and material handling system design is required to achieve the benefits of improved fab performance and justify the large investment costs of a semiconductor wafer fabrication facility.

In this paper, a spacefilling curve (SFC) heuristic is proposed to solve the layout and material handling system design integration problem. SectionI I reviews the literature pertinent to this integration design problem. The details of the proposed procedures for the spine and perimeter configurations are discussed in sections III and IV, respectively. Section V contains empirical illustrations of the proposed approach. Conclusions and suggestions for future research are given in sectionV I.

II. LITERATURE REVIEW

Most of the literature related to design of semiconductor fabrication facilities has concentrated on the material handling system, e.g., [1] and [2]. Most of these approaches have used simulation models to analyze alternative material handling system designs given a fab layout2 []. These approaches do not provide information on how to create a good design or how to integrate the design of the layout with the material handling system.

There has also been previous research investigating facility layout and material handling system design in applications not specific to semiconductor manufacturing. For example, Dowlatshahi6 [] and Lacksonen [7] both provide broad frameworks for integrating manufacturing system design issues. However, detailed design procedures are not provided in these frameworks.

Montreuil [8] proposed a modeling framework for integrating layout and undirected flow network designs. Chhajed *et al.* [9] provided a detailed flow network design on an existing layout. Langevin *et al*. [10] solved a spine layout design problem with an undirected material flow network. Their procedure searches for a rectangular layout shape along a central spine. While this method is effective at solving many manufacturing system design problems, it is not easily applied to the fab design problem because its output must be modified to fit the fab design criteria. These modification requirements, and their potential detrimental effect on performance, are due to two major reasons. First, the design objective considers an undirected flow network, which is not typical of AMHS used in semiconductor fabs. Second, it does not enforce the cell geometry and exact P/D point positions, which are necessary and important components of the fab design problem.

Given a fixed single-loop material flow path, Wu and Egbelu 1[1] developed a procedure to determine an optimal layout design along this path. Banerjee and Zhou 1[2] designed a directed, single loop machine layout by sequentially determining the flow sequence between machines and the layout of the machines. Yang and Peters [13] used a spacefilling curve approach to solve a facility layout design problem in an FMS with a simple material flow loop configuration. They then add shortcuts to the material handling loop using a one at a time procedure to attempt to reduce the material handling cost. All of these methods could potentially be applied to the semiconductor fab design problem. However, each would require manual adjustment to the resulting solution, as discussed above. Thus, the ultimate quality of the resulting layout would be suspect. Furthermore, none of these procedures exploit the structure of the fab design problem, which allows a specialized procedure to be developed that is both effective and efficient at determining the fab layout and material handling system design.

Tong [14] and Tate and Smith [15] developed a sequential construction technique and a genetic search heuristic, respectively, to solve a flexible bay structure layout, which is similar to the perimeter bay configuration shown in Fig. 2. This design allows a bay to contain more than two departments; hence, it is possible, when this procedure is applied to a perimeter configuration semiconductor fab, to have a department that cannot interface with the AMHS since it doesn’t have a boundary on the perimeter of the layout.

The current paper considers a directed material flow path but it also exploits the special structure of a semiconductor fab configuration. It differs from previous literature in that it integrates the layout and material handling system design by considering shortcuts that result from the addition of crossover turntables and generates a bay layout design that is customized for a semiconductor fabrication facility.

III. SPINE CONFIGURATION

A. Overview

When modeling the integrated design problem with a spine configuration, three interesting aspects are noted. First, the material handling system can be modeled as a single loop with crossover points. Second, the bay layout arrangement with a centralized spine material handling system requires that each department have exactly one boundary on the center spine of the floor space in order to access the material handling system, which implies that exactly one department can be located in each bay.

Third, the bay height is usually constant across the entire length of the facility. Typically, the bay height will be the same in the upper row and the lower row, which implies that the material handling system is in the center of the facility. However, the procedures described in this paper allow the upper and lower row heights to differ. We will assume, however, that these heights are design parameters that are based on the desired bay configurations and available floor space and, hence, are specified prior to solving the integrated layout problem.

B. Spacefilling Curve Formulation

Spacefilling curves (SFC) have been used to develop efficient heuristics to solve several types of combinatorial problems such as traveling salesman, routing, and partitioning problems1 [6] − [18], and more recently layout problems [19], [13].

The typical SFC approach divides the available floor space and departments into small equal-size square blocks. When this approach is applied to the fab layout, the floor space is divided into *equal area* *unit rectangular blocks*. Each block extends the full height of the bay, with the width of the unit blocks depending on the height of the row. Each block is assigned a distinctive value as its address.F ig. 3 illustrates the address assignment, which is used to determine the set of blocks that are occupied by a department. Note that, for modeling convenience, the numbering of the address assignments follows the AMHS flow direction with the upper-left corner block as the starting block.

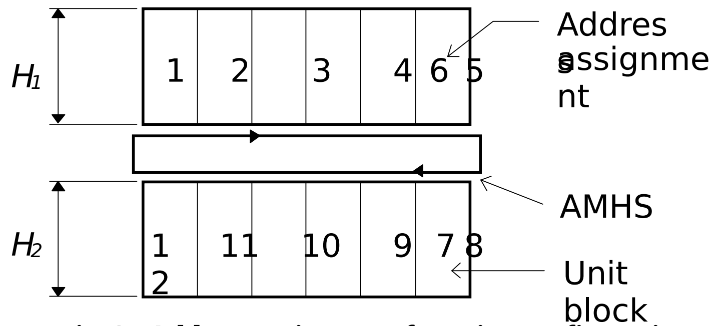
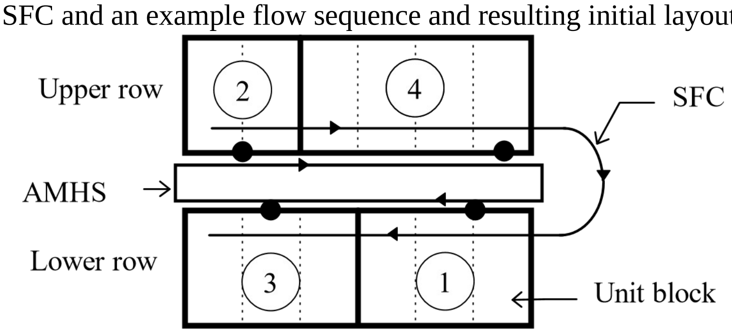


Fig. 3. Address assignment for spine configuration.

The position of a pickup/dropoff (P/D) point (typically a stocker) of a department is represented by its distance from the starting block that defines the department boundary; hence, it is the distance from the lefthand side of the starting block when the department is located in the upper row, while it is the distance from the right-hand side of the starting block when the department is located in the lower row. Obviously, the P/D point is located on the department boundary adjacent to the material handling system.

When the SFC is applied to the spine configuration, there is a unique SFC that is coincident with the AMHS flow loop. Based on the SFC, an initial flow sequence is specified and used to create a layout.

Fig. 4 shows the SFC and an example flow sequence and resulting initial layout.



Flow sequence: 2→4→1→3

Fig. 4. SFC example for spine configuration.

Since the flow sequence is not unique, multiple layouts can be generated and evaluated. The user can then determine the set of flow sequences and layouts to consider further, which is a similar idea to other layout procedures including CRAFT [20], ALDEP [21], and MULTIPLE [19].

Once the SFC and initial flow sequence have been determined, the material handling system design determines the number of shortcuts to be added to the layout and the location for each shortcut. This material handling system design problem is modeled using a network flow formulation (as described in the next section) to minimize the trade-off between the increase in shortcut investment costs and the decrease in material handling costs. The procedures for determining the layout arrangement and material handling system design are embedded into an iterative steepest-descent-pairwise-interchange (SDPI) procedure for solving the overall integrated design problem.

C. Crossover Turntable Design

III.C.1 Background

The addition of a crossover turntable forms a shortcut on the AMHS loop and reduces material handling costs. Crossover turntables are added to the single-loop material flow path, which forms the core of the wafer fab with a spine configuration. The design of the number and location of the crossover turntables depends on the available floor space, control strategy employed, acquisition and installation costs, *etc*.

In general, there is a tradeoff between the additional costs associated with installing a crossover turntable and the resulting decrease in material handling costs. Therefore, additional crossover turntables will be installed as long as the incremental cost of adding the turntable is offset by the savings in material handling cost. The following subsections present a network flow formulation for determining the optimal number and location of crossover turntables given a layout.

III.C.2 Modeling of the Crossover Turntable Design Problem

In practice, there are two design constraints for installing a crossover turntable. First, there must be a minimum clearance between the crossover turntable and the nearest P/D point, as well as between two crossover turntables. This clearance depends on the particular material handling system technology and will be denoted by *e*. In addition, the crossover turntable uses the shortest path to divide the AMHS loop as illustrated in Fig. 1, which implies that the crossover turntable rail is perpendicular to the material handling rail.

The proposed design procedure first sorts the P/D points in the horizontal direction beginning from the left boundary of the floor space so that the positions of P/D points can be transformed from a twodimensional space to a single dimension. For example shown inF ig. 4, the P/D point of department 2 when projected to the horizontal line, is 1.0 distance unit from the left boundary of the floor space. Similarly, the P/D point for departments 3, 1, and 4 are 1.5, 5.0, and 5.5 distance units, respectively. The resulting projections are shown in Fig. 5.

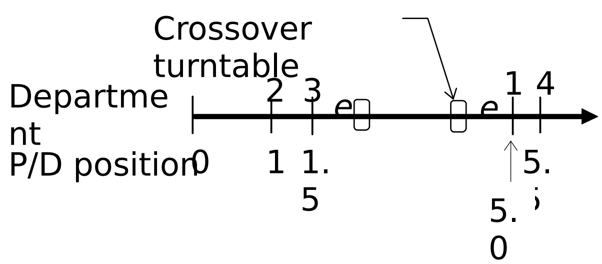


Fig. 5. P/D point sorting example.

In this example, there are three candidate intervals to locate the crossover turntable: one between departments 2 and 3, one between 3 and 1, and the other between departments 1 and 4 The candidate locations of the crossover turntables inside an interval can be restricted by considering the nature of the material flow. Consider an interval between two P/D points A and B lying on a clockwise material flow path and assume that wafer lot boxes are transported on the shortest path. Any lots originating from departments on the raise row to the right of B with destinations in the upper row to the right of B will pull the crossover turntable as close to B as possible to minimize the travel distance. Similarly, loads originating in the upper row to the left of A that are destined for the lower row to the left of A will pull the crossover turntable as close to A as possible.

If two turntables are located in the interval with one next to B and with one next to A, then no other turntable located between these two will ever lie on the shortest path between any origin/destination pear, and hence this turntable will never be used. Therefore, if a shortcut is added, it will be as close to a P/D point as possible. Due to the minimum clearance restrictions, the candidate crossover turntable design for an interval with length Δ has three conditions:

1. If $\Delta < 2e$, then it is not feasible to install a crossover turntable in this interval and therefore the interval is excluded from further consideration.

2. If $2e \leq \Delta < 3e$, then the interval has two candidate locations but can contain at most one crossover turntable.

3. If $\Delta \geq 3e$, then there are two candidate locations for crossover turntables and both can be selected.

Each candidate location for the installation of a crossover turntable adds two vertices to the flow network as illustrated in Fig. 6 (from the example in Fig. 4). In this figure, the shortcut under consideration is located in the interval between departments 3 and 1. Note that vertices 5, 6, 7, and 8 are pseudo vertices representing the crossover turntable position.

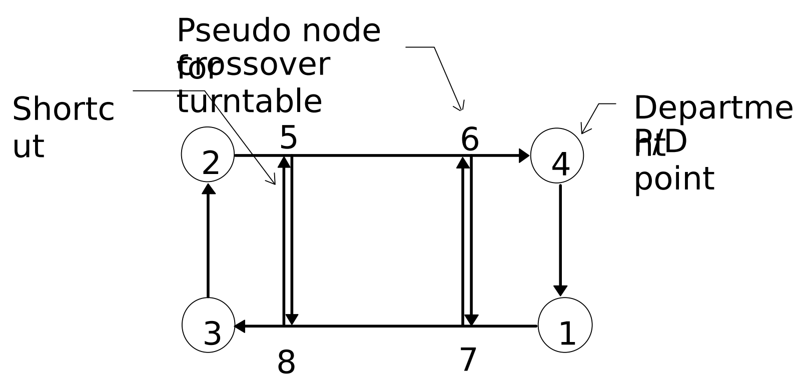


Fig. 6. Shortcut configuration example.

If vertices (V) represent the P/D points of departments and the pseudo nodes of crossover turntables in the layout, and edges (E) represent directed flow segments between vertices, then the standard network representation $G = (V, E)$ can be used to describe the problem.

Determining the subjective function is usually a challenge for the facility design engineer due to the potential conflicting objectives and other factory-wide design factors. Furthermore, the operational design issues such as scheduling strategies, product types, and production demands, all impact the layout and material handling system design objective but are very difficult to predict exactly at the design stage. Therefore, we follow the common practice of minimizing the weighted flow distance for layout and material handling system design problem. In the following discussions, "cost" will be used in the objective function although it is not necessary to be a dollar unit. Developing a design that can efficiently handle the anticipated production requirements should enable the material handling system to effectively respond to dynamic operation requests.

Given a feasible AMHS flow path configuration and the candidate shortcuts, the length of each arc is known. Let δ_{ij} be the directed flow distance of an arc (i, j). Define a directed flow density from the P/D point of a department to the P/D point of another department as a commodity k, then the set of directed flow densities is K with $k \in K$. Let u^k be the cost per unit distance for routing commodity k, and c_{ij}^k be the cost per unit for routing commodity k on the directed arc (i, j). Then, $c_{ij}^k = u^k \delta_{ij}$.

Let F be the cost per period for the installation of a crossover turntable. Note that the per-period-cost not only considers the system acquisition costs, but may also include operating costs, and is discounted to reflect the useful life of the material handling system. Other system costs are assumed independent of the layout and material handling system design and are therefore not included in the analysis. The following additional notation is needed to model the problem.

A_i : the area of department i
H_1: the height of the upper row (as illustrated inF ig. 3)

H_2: the height of the lower row (as illustrated inF ig.

3) L : the length of the complete AMHS loop a : the

fixed distance across the center spine x_{ij}^k: the flow of

commodity k on the directed arc (i, j)

y_{ij}: the crossover turntable (shortcut) arc (i, j)

$D(k)$: the dropoff point (sink node) of commodity k

$P(k)$: the pickup point (source node) of commodity k

$\bar{S}$: the set of directed shortcut arcs (connecting pseudo nodes), $\bar{S} \subset E$

S: the set of undirected shortcut arcs (connecting pseudo nodes), $S \subset \bar{S}$

The width of a unit-rectangular block in the upper row is assumed to be 1. This width can be easily achieved by normalizing the scale. Then, the width of the unit-rectangular block in the lower row is $b = H_1/H_2$. Note that L, A_i, a, and b are used to calculate the arc length δ_{ij}, which is a straightforward calculation given these parameters and the layout arrangement of departments in bays. Then, the integrated shortcut design is formulated as equations (1) − (5).

$$\text{Minimize} \qquad Z = \sum_{k \in K} \sum_{(i,j) \in E} c_{ij}^k x_{ij}^k + \sum_{(i,j) \in S} F y_{ij} \tag{1}$$

$$\text{st: } \sum_{j \in V} x_{ji}^k - \sum_{j \in V} x_{ij}^k = \begin{cases} -1 & \text{if } i = D(k) \\ 1 & \text{if } i = P(k) \\ 0 & \text{otherwise} \end{cases} \qquad \forall i \in V \text{ and } \forall k \in K \tag{2}$$

$$x_{ij}^k,\ x_{ji}^k \le y_{ij} \qquad \forall (i, j) \in S \text{ and } \forall k \in K \tag{3}$$

$$x_{ij}^k \ge 0 \qquad \forall (i, j) \in E \text{ and } \forall k \in K \tag{4}$$

$$y_{ij} \in \{0, 1\} \qquad \forall (i, j) \in S \tag{5}$$

Given a flow sequence, equation 1() optimizes the trade-off between the increased crossover turntable investment costs and the decreased material handling costs. Equations 2() are the conservation of flow constraints for each commodity k. The *forcing* constraints, equation (3), state that if $y_{ij} = 0$, *i.e.*, if shortcut (i, j) is not included in the design, then the flow of every commodity$k$ on this arc must be zero in both directions. This formulation is a multi-commodity network flow problem with the addition of fixed charge variables for certain arcs. The network properties will be exploited in efficiently determining a solution.

D. Integrated Design Improvements

A steepest-descent-pairwise-interchange (SDPI) heuristic is used to improve the solution by evaluating the cost reduction resulting from exchanging two departments in the flow sequence. The procedure iterates until there is no further improvement. Note that since a department cannot split across the center spine into both the upper and lower rows, some flow sequence solutions will be excluded from further consideration since they would result in infeasible layout designs.

Each pairwise interchange of departments results in a new flow sequence, which leads to a new layout, which in turn results in a new layout cost. Let S_i and O_i be the starting block and P/D point position for department i, respectively. For the example shown in

Fig. 4 (assume that b = 1 and there are no shortcuts), $\delta_{13} = (S_3 - S_1 - O_1 + O_3)$. Since a directed material flow path is being used, if departments 3 and 1 are interchanged, $\delta_{13} = L - (S_1 - S_3 + O_1 - O_3)$. This pairwise exchange heuristic has been effective in solving quadratic assignment problems. Readers are referred to Francis *et al*. [22] for a more detailed discussion of the basic SDPI heuristic.

Using these insights and results described above, the overall procedure for determining the integrated design is shown in Fig. 7, where N is the total number of departments to be included in the layout.

IV. PERIMETER CONFIGURATION

A. Overview

In this section, the SFC formulation from sectionI II.B is modified to solve the perimeter configuration design problem. With a perimeter configuration, there are several modeling aspects that must be considered. First, the material handling system can be modeled as a double-loop with crossover points. In addition, the bay layout arrangement with a perimeter material handling system requires that each department have at least one boundary on the perimeter of the floor space in order to access the material handling system, which implies that at most two departments can be located in any bay. Finally, the requirement of the maintenance spaces between bays dictates that the bay width must be constant across the entire width of the facility. These aspects are illustrated inF ig. 2.

Thus, the problem becomes one of determining the number of bays, the width of each bay, the departments assigned to each bay, the location of the P/D points, and the number and location of crossover turntables on the material handling system.

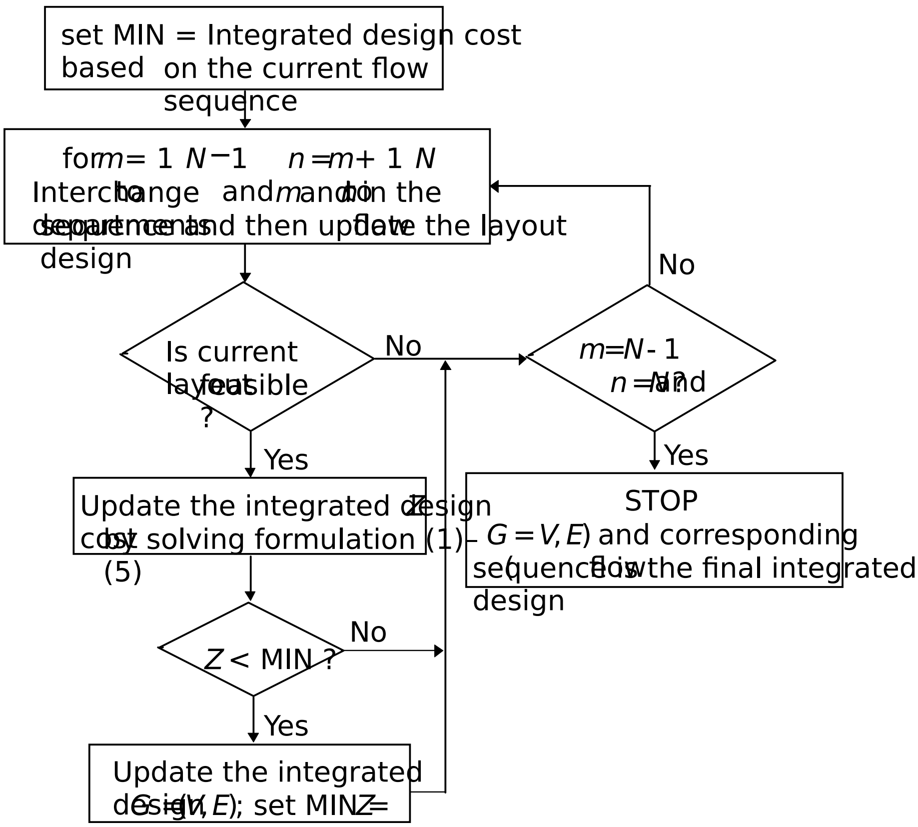


Fig. 7. Integrated design improvement procedure.

B. SFC Formulation

The SFC formulation divides the bay floor space and departments into equal size unit rectangular blocks as illustrated in Fig. 8. The SFC in this case is just a line from left to right. For a general design in practice, there will be two departments in a bay; hence, a *pseudo* department, which pairs two departments together, is proposed to fill the floor space following the SFC pattern. Given an initial flow sequence, *e.g.*, $1,\ldots,n$, the first pseudo department consists of the first and last departments in the sequence, *i.e.*, 1 and n, the second pseudo department consists of the second and next-to-last departments, *i.e.*, 2 and n–1, the third pseudo department consists of departments 3 and n–2, and so forth. When there is an odd number of departments, the last included department, which is in the middle of the sequence, will occupy a bay by itself.

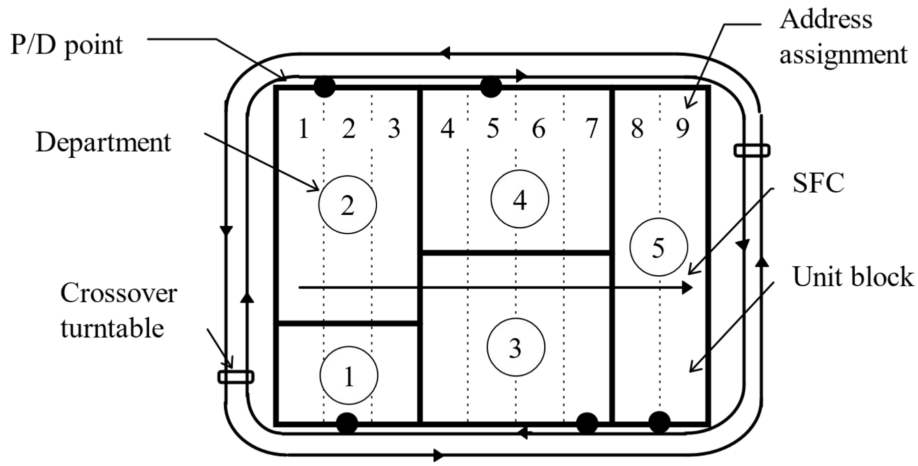


Fig. 8. Address assignment for perimeter bay configuration.

Let $A = [A_i, i=1,..,n]$ represent department areas in terms of number of unit rectangular blocks. For the example in Fig. 8, the department area matrix is $A = (1, 2, 2, 2, 2)$. If the initial flow sequence is (2→4→5→3→1), then the pairing results in three pseudo departments which have departments 2 and 1 as the first pseudo department, departments 4 and 3 as the second pseudo department, and finally department 5 as the third pseudo department. Let A' be the area matrix of the pseudo departments, then $A' = (3, 4, 2)$ for this example. The pseudo departments then fill the floor space according to the SFC. If a pseudo department is comprised of two departments, then the boundary between these two departments is determined by the ratio of their areas. For example, in pseudo department 1, the bay is divided into two regions with a 2 to 1 ratio in area for departments 2 and 1, respectively. This procedure creates a flow sequence and then places departments in the layout so that this flow sequence is maintained, given the perimeter-based material handling system.

C. Crossover Turntable Design

The problem of locating a crossover turntable can be transformed to a standard network flow formulation and solved efficiently. Since a department's P/D point can only interface with the inner AMHS loop, which we define as circuit I, any flow must move back to this circuit in order to pickup or dropoff wafer lot boxes at a department.

Thus, the double-loop, perimeter AMHS configuration can be mapped to two separate circuits as illustrated in Fig. 9 for the example from Fig. 8. At least one crossover turntable is required to use the outer loop (circuit II). The addition of a crossover turntable adds an arc (β β ’) to connect the two circuits as shown in Fig. 9. For the outer loop to be useful, at least one additional crossover turntable is needed to connect the inner and outer loop, *e.g*., arc (γ, γ’). Otherwise, a vehicle can get on the outer loop but must complete the entire circuit and return to the crossover in order to get back on the inner loop, which puts the vehicle exactly back where it started. Therefore, in general there will be two or more crossover turntables associated with the double-loop, perimeter AMHS configuration. The flow path from the P/D point of a department to the P/D point of another department will be the shortest path between the two vertices representing the actual pickup/deposit points (not the pseudo vertices used in circuit II, although they may be intermediate vertices in the path).

The determination of the number and location of candidate shortcuts is carried out as discussed in section III.C with the integrated design problem formulated as a network flow problem so that it can be solved efficiently.

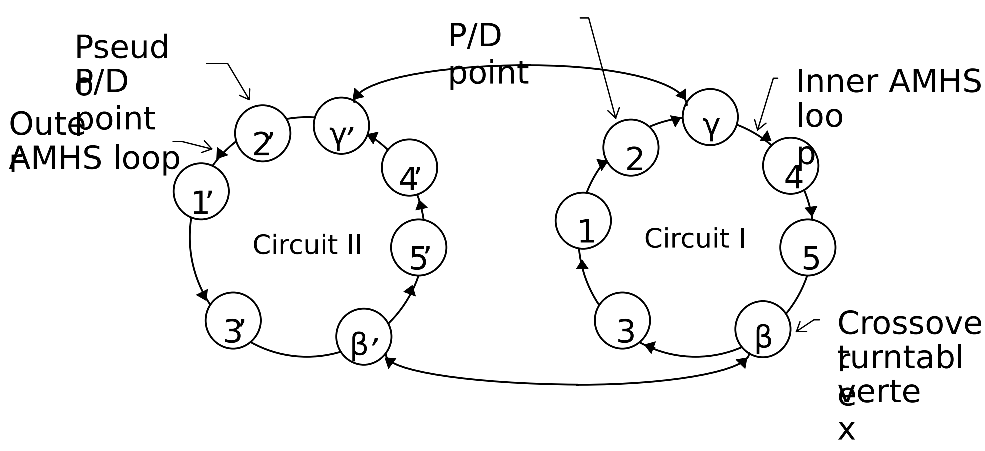


Fig. 9. Flow network for perimeter configuration.

D. Integrated Design Improvements

Every potential flow sequence results in a potential layout, with an objective function value representing the material handling cost. The SDPI heuristic discussed in sectionI II.D is used to improve an initial flow sequence and search for the best solution. Note that after each interchange, the flow sequence and pseudo departments must be updated in order to compute the resulting material handling cost.

The integrated shortcut design formulation is the same as equations 1() − (5) and the improvement procedure is as shown in Fig. 7 for the spine configuration, although now the calculation of the material handling distance is using the two circuit network shown inF ig. 9. Therefore, the details of the design procedures are not repeated here.

V. EMPIRICAL ILLUSTRATIONS

In order to demonstrate the proposed procedures, three example problems, with 8-, 14-, and 20-bays, are generated using the data in Carpenter *et al*. [2] as a guide. The area information and inter-department flow densities are as shown in Appendices 1, 2, 4, and 5. It is assumed that the minimum distance between a crossover turntable and a P/D point (or another crossover turntable) is equal to 0.6 distance units.

The proposed procedures are coded in C^{++} and implemented on an IBM RS/6000 computer. Each integrated shortcut design formulation is solved using a combined network-dual simplex method optimizer provided by CPLEX® [23] in a callable library.

A. Spine Configuration

For the spine configuration, additional assumptions include: (1) both H_1 and H_2 are equal to 5 distance units; (2) the discounted fixed cost per period for a crossover turntable is 70; and (3) the length between the two bay rows in the spine configuration is 2 distance units.

Since the initial flow sequence impacts the solution quality for the SDPI procedure, each problem is repeated five times with different initial flow sequences as shown in Appendix 3. The experimental results are summarized in Table 1. The second column shows the lowest cost obtained from the five runs. The third and fourth columns are included to illustrate the dependency of the algorithm on the initial flow sequence. As can be seen, the average is slightly higher, but the algorithm is able to achieve a good solution from a variety of starting sequences, which is important for this type of problem.F ig. 10 illustrates one of the integrated designs determined by the procedure.

The solution quality of this proposed integrated design procedure is demonstrated by comparing it with the solution from an exact layout design formulation, a modified quadratic set covering formulation (QSCP). The modified QSCP formulation is capable of optimally solving the fab layout design problem with a simple flow path; however, it is computationally prohibitive for a realistic size problem. Readers are referred to Peters and Yang [24] for a detailed discussion on the QSCP formulation in the context of a more general layout design problem.

Table 1 . Spine configuration costs.

$n = 5$	Minimum cost	Average cost	Std. Deviation
8-bay	5613.5	5659.9	38.9
14-bay	17705.4	18539.2	828.7
20-bay	46878.6	47797.1	982.1

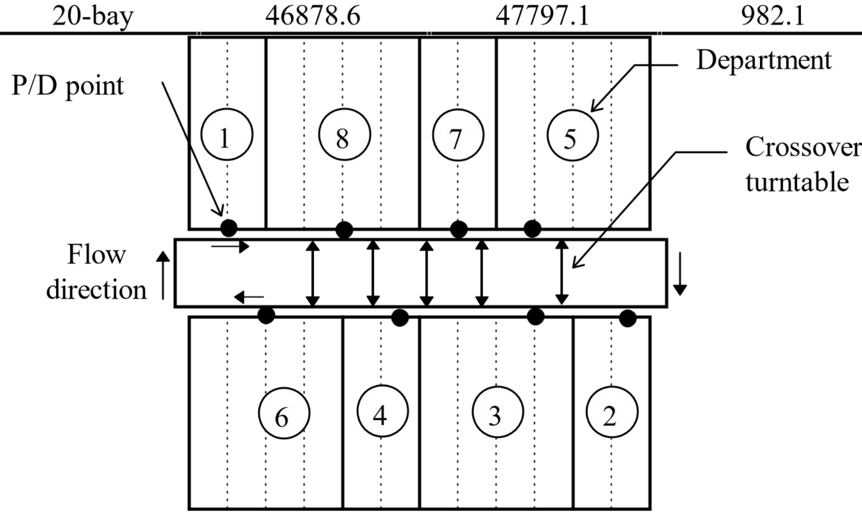


Fig. 10. Spine configuration integrated design.

Once the optimal layout design is found using the modified QSCP formulation, the shortcut design is then solved by using equations (1) – (5). For the 8-bay spine configuration problem, the resulting integrated design cost by using the modified QSCP formulation is *5737.5* which is higher than *all* of the costs from the five repetitions of the SFC procedure. Clearly, integrating the consideration of the layout and material handling system design yields benefits relative to a sequential solution approach even if an optimal layout procedure is used.

Clearly, there is a tradeoff between the investment cost of additional crossover turntables and the reduced material handling costs. The integrated design procedure will determine the best solution given the fixed crossover turntable cost. To illustrate this relationship, the fixed cost was varied and the resulting solution in terms of the material handling cost and total cost as well as the number of shortcuts was determined. These results are shown graphically for the 8-bay problem inF ig. 11 and Fig. 12. The procedure can be used to provide this type of sensitivity analysis in situations where there is uncertainty in the additional cost associated with installing a crossover turntable.

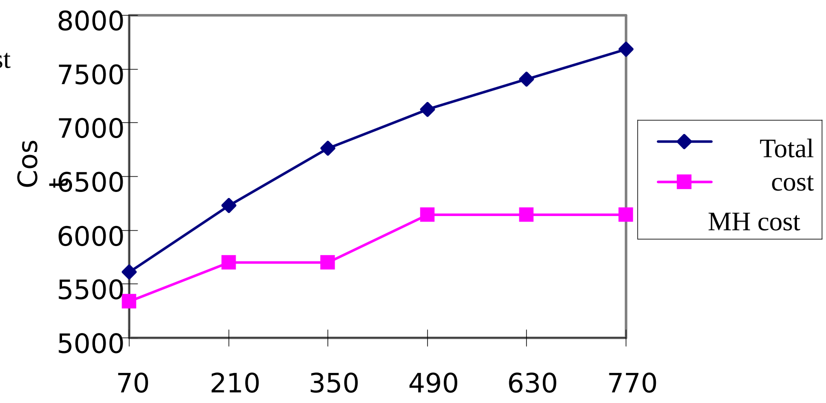


Fig. 11. Sensitivity analysis of fixed crossover turntable cost.

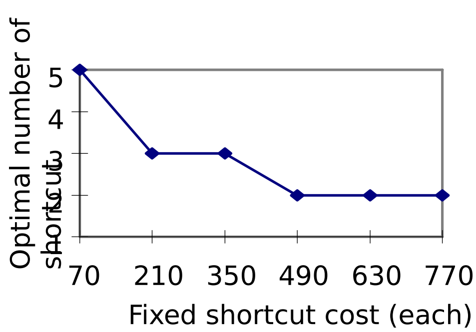


Fig. 12. Fixed cost versus number of crossover turntables.

B. Perimeter Configuration

For the perimeter configuration, the examples assume that: (1) the bay length is equal to 10 units for a perimeter configuration; (2) the crossover turntable cost is 60, and (3) the length of crossover turntable is 0.5, which is smaller than the length of 2.0 for a spine configuration and contributes to the lower cost.

Each problem size is repeated five times with different initial flow sequences as shown in Appendix 6. Table 2 shows the integrated design costs andF ig. 13 illustrates one of the integrated designs that were obtained. Notice that the cost of the perimeter configuration is significantly higher than the cost of the spine configuration for the corresponding problem.

Table 2 . Perimeter configuration costs.

$n = 5$	Minimal cost	Average	Std. Deviation
8-bay	9347.3	9551.9	241.1
14-bay	29827.8	30522.0	859.1

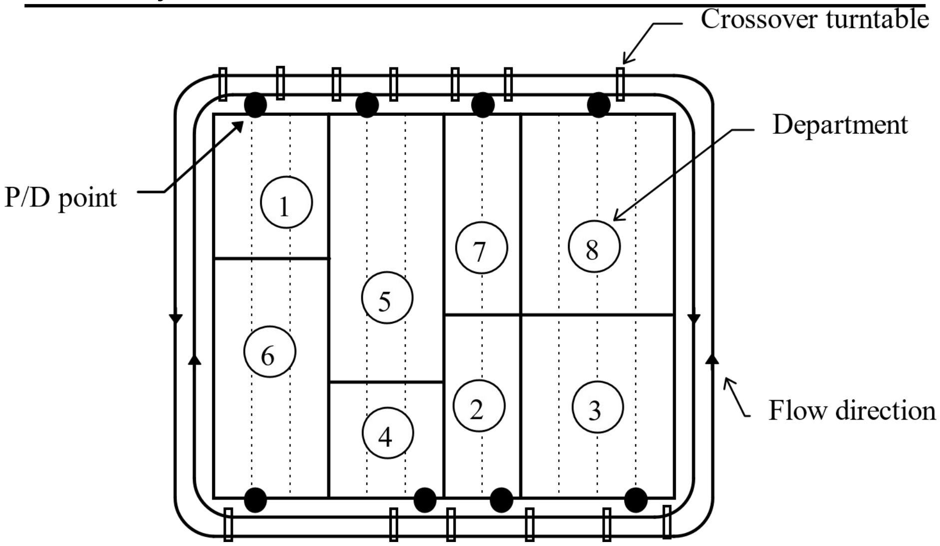


Fig. 13. Perimeter configuration integrated design.

VI. CONCLUSIONS

This paper presents a combined spacefilling curve and network flow procedure that solves the integrated layout and material handling system design problem efficiently and effectively for both the spine and perimeter configurations in semiconductor fabrication facilities. The special structure of semiconductor fabs was exploited in developing the procedures. Experimental results with the procedure were presented and are very encouraging.

Since the bay configuration with an overhead monorail material handling system is the trend for future semiconductor facility designs, the procedure developed in this paper is an important step toward efficient and effective semiconductor fab design in order to improve the system performance in terms of throughput, yield increases, WIP reduction, tool utilization, *etc*. In addition, this integrated design procedure can be adapted to the intrabay integrated design problem, which is a micro-version of the interbay design problem. Therefore, the integrated design procedure is applicable to real world problems facing the semiconductor industry.

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APPENDICES

1. Department area information for 8-bay problem

Department	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Area (x5)	2	2	4	2	4	4	2	4
P/D offset	1	0.5	1	0.5	1	2	1	2

2. From-to departmental flow matrix for the 8-bay problem

From\To	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
1	0	10	4	20	19	1	19	18
2	16	0	10	16	4	16	5	1
3	13	9	0	10	12	16	15	9
4	16	19	10	0	14	15	8	4
5	17	17	10	8	0	8	20	12
6	17	17	10	8	0	8	20	12
7	4	20	19	1	19	18	0	6
8	10	16	4	16	5	1	10	0

3. Flow sequences for 8-bay problem

Repetition No.	**Flow sequences for SFC**
1	1 → 5 → 7 → 8 → 2 → 3 → 4 → 6
2	7 → 6 → 1 → 8 → 2 → 3 → 4 → 5
3	2 → 3 → 4 → 5 → 1 → 6 → 7 → 8
4	4 → 5 → 6 → 7 → 1 → 2 → 3 → 8
5	5 → 6 → 8 → 7 → 4 → 3 → 2 → 1

4. Department area information for 14-bay problem

Department	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**
Area (x5)	2	4	2	4	2	4	2	2	2	6	2	4	2	2
P/D offset	1.0	1.0	1.0	1.0	1.0	0.5	1.0	1.0	2.0	2.0	1.0	0.5	0.5	2.0

5. From-to departmental flow matrix for the 14-bay problem

From\To	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**
1	0	16	1	11	15	13	14	10	19	15	1	12	9	8
2	6	0	12	3	6	4	19	11	7	0	15	12	8	6
3	7	2	0	9	9	3	6	1	10	18	5	19	10	13
4	13	9	5	0	13	17	16	8	0	18	16	0	2	2
5	7	15	16	18	0	19	8	19	1	17	15	1	15	0
6	10	1	17	15	3	0	12	11	13	13	14	13	6	18
7	6	10	4	14	11	19	0	10	4	11	12	15	3	6
8	4	0	8	4	19	8	5	0	15	0	3	0	18	16
9	7	4	8	0	13	5	9	3	0	12	5	11	19	2
10	4	11	0	10	1	16	3	5	2	0	13	18	12	1
11	1	1	12	2	2	14	18	5	9	3	0	16	5	5
12	19	3	2	6	9	13	19	7	10	6	3	0	8	17

13	13	6	15	7	14	5	18	3	3	14	19	11	0	1
14	19	11	3	18	14	4	16	13	7	7	12	12	6	0

6. Flow sequences for 14-bay problem

Repetition No.	Flow sequences for SFC
1	10 → 11 → 12 → 13 → 14 → 4 → 9 → 8→ 7 → 6 → 5 → 3 → 2 → 1
2	1 → 2 → 3 → 7 → 8 → 9 → 10 → 4→ 5 → 6 → 11 → 12 → 13 → 14
3	6 → 7 → 8 → 9 → 10 → 12 → 14 → 13→ 11 → 5 → 4 → 3 → 2 → 1
4	2 → 4 → 5 → 8 → 9 → 10 → 14 → 13→ 12 → 11 → 7 → 6 → 3 → 1
5	10 → 2 → 4 → 5 → 6 → 14 → 13 → 12→ 11 → 9 → 8 → 7 → 3 → 1

7. Department area information for 20-bay problem

Department	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Area (x5)	2	2	2	4	2	6	4	2	2	4	4	2	2	2	4	2	6	2	4	2
P/D offset	1.0	1.0	1.0	1.0	1.0	0.5	1.0	1.0	2.0	2.0	1.0	0.5	0.5	1.0	1.0	1.0	1.0	0.5	0.5	0.5

8. From-to departmental flow matrix for the 20-bay problem

From\To	1	2	3 6	4 4	5 12	6 3	7 6	8 4	9 19	10 11	11 7	12 0	13 15	14 12	15 8	16 6	17 7	18 2	19 14	20 9
1	0	8																		
2	9	0	6	1	10	18	5	19	10	13	13	9	5	2	13	17	16	8	0	18
3	16	0	0	2	7	15	16	18	16	19	8	19	1	17	15	1	15	0	10	1
4	17	15	3	0	12	11	13	13	14	13	6	18	6	10	4	14	11	19	6	10
5	4	11	12	15	0	6	4	0	8	4	19	8	5	6	15	0	3	0	18	16
6	7	4	8	0	13	0	9	3	0	12	5	11	19	2	4	11	0	10	1	16
7	3	5	2	12	13	18	0	1	1	1	12	2	2	14	18	5	9	3	12	16
8	5	5	19	3	2	6	9	0	19	7	10	6	3	12	8	17	13	6	15	7
9	14	5	18	3	3	14	19	11	0	1	19	11	3	18	14	4	16	13	7	7
10	12	12	6	2	5	13	5	8	10	0	4	16	18	4	6	16	14	1	9	3
11	6	2	0	15	0	8	15	12	17	11	0	4	8	11	18	1	11	3	16	11
12	2	9	15	18	10	15	2	15	16	2	6	0	15	19	16	16	3	3	9	9
13	2	9	1	0	5	9	10	1	11	6	3	11	0	3	0	4	10	18	7	1
14	16	6	19	6	5	19	11	5	1	2	6	14	13	0	5	19	14	1	2	12
15	5	8	0	13	17	1	7	6	8	9	16	3	14	16	0	9	12	0	16	0
16	4	18	1	14	4	3	14	13	0	7	10	12	8	10	4	0	19	12	15	7
17	6	18	3	11	3	1	16	9	4	2	5	0	3	7	9	11	0	12	8	10
18	5	5	13	7	4	7	7	7	13	4	9	5	15	8	2	11	14	0	0	8
19	18	10	4	14	18	12	15	4	1	16	19	2	0	9	4	19	13	16	0	8
20	16	8	13	7	11	13	10	18	7	15	12	5	17	12	10	0	17	5	19	0

9. Flow sequences for 20-bay problem

Repetition No.	Flow sequences for SFC
1	1 → 2 → 3 → 7 → 8 → 9 → 10 → 11→ 18 → 19 → 20 → 4 → 5 → 6→ 12 → 13 → 14 → 15 → 16 → 17
2	14 → 15 → 16 → 17 → 18 → 19 → 6 → 7→ 20 → 13 → 12 → 11 → 10 → 9→ 8 → 5 → 4 → 3 → 2 → 1
3	2 → 3 → 4 → 10 → 11 → 12 → 15 → 17→ 20 → 19 → 18 → 16 → 14 →

	13→ 9 → 8 → 7 → 6 → 5 → 1
4	3 → 4 → 7 → 8 → 9 → 12 → 15 → 17→ 19 → 20 → 18 → 16 → 14 → 13→ 11 → 10 → 6 → 5 → 2 → 1
5	6 → 7 → 10 → 11 → 12 → 15 → 17 → 20→ 19 → 18 → 16 → 14 → 13 → 9→ 8 → 5 → 4 → 3 → 2 → 1

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Brett A. Peters
Department of Industrial Engineering
Texas A&M University
College Station, TX 77843-3131

(409) 845-3574
(409) 847-9005
bpeters@tamu.edu

Brett A. Peters is an Assistant Professor in the Industrial Engineering Department at Texas A&M University. He received a Ph.D. and M.S. in Industrial Engineering from the Georgia Institute of Technology in 1992 and 1988, respectively, and a B.S.I.E. from the University of Arkansas in 1987. His research and teaching interests include facilities design, material handling systems, and the design, analysis, operation, and control of manufacturing systems. Dr. Peters is a member of SME, IIE, and INFORMS.

Taho Yang
TEFEN USA
1065 E. Hillsdale Blvd Suite #400
Foster City, CA 94404
PH: 415-577-8094

Taho Yang was born in Keelung, Taiwan on July 28, 1963. He received the Ph.D. degree from the Industrial Engineering Department at Texas A&M University in 1996. Then, he joined Tefen USA as an Industrial Engineer. His primary research interests include design and control of manufacturing systems with emphasis on semiconductor fabrication facilities, and the integrated layout and material handling system design problems.